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Section: BCS-3K

Course: Computer Organization and  
Assembly Language – Lab

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Lab: Lab 6

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Task 1:

INCLUDE Irvine32.inc

.data

array DWORD 5 DUP(?)

msg1 BYTE "Enter Quiz Score: ", 0

msg2 BYTE "The Total Sum is: ", 0

sum DWORD 0

.code

main PROC

mov ESI, 0

mov ECX, LENGTHOF array

L1:

mov EDX, OFFSET msg1

call WriteString

call ReadInt

add sum, EAX

mov array[ESI\*TYPE array], EAX

inc ESI

call Crlf

loop L1

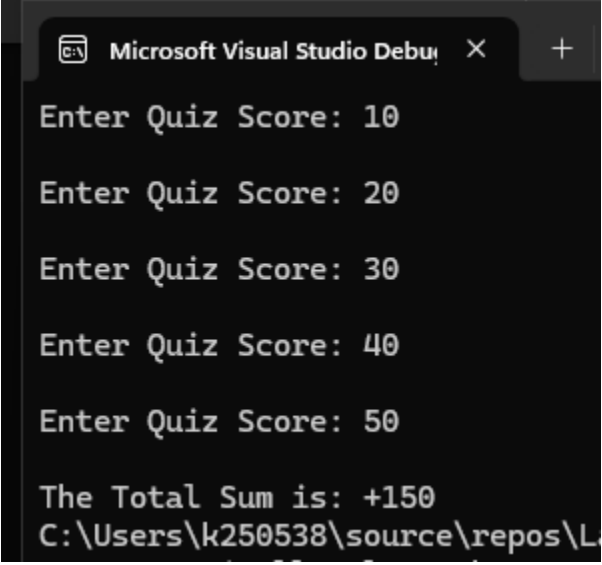
mov EAX, sum

mov EDX, OFFSET msg2

call WriteString

call WriteInt

```
exit
main ENDP
END main
```



The screenshot shows a Windows-style window titled "Microsoft Visual Studio Debug Console". The window has a dark background with light-colored text. The text inside the window is as follows:

```
Enter Quiz Score: 10
Enter Quiz Score: 20
Enter Quiz Score: 30
Enter Quiz Score: 40
Enter Quiz Score: 50

The Total Sum is: +150
C:\Users\k250538\source\repos\L
```

Task 2:

```
INCLUDE Irvine32.inc
```

```
rows = 3
```

```
columns = 4
```

```
total_elements = rows * columns
```

```
.data
```

```
records DWORD 10, 15, 20, 25
```

```
        DWORD 30, 10, 5, 15
```

```
        DWORD 20, 20, 20, 20
```

```
msg1 BYTE "The Total Deliveries made are: ", 0
```

```
.code
```

main PROC

mov ESI, 0

mov EAX, 0

mov ECX, total\_elements

L1:

add EAX, records[ESI \* TYPE records]

inc ESI

loop L1

mov EDX, OFFSET msg1

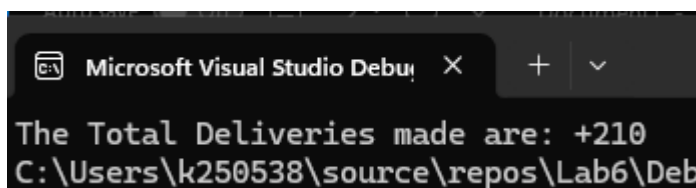
call WriteString

call WriteInt

exit

main ENDP

END main



Task 3:

```
INCLUDE Irvine32.inc
```

```
.data
```

```
.code
```

```
main PROC
```

```
mov EAX, 49
```

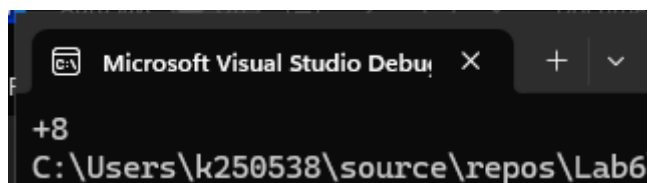
```
call RandomRange
```

```
call WriteInt
```

```
exit
```

```
main ENDP
```

```
END main
```



Task 4:

INCLUDE Irvine32.inc

.data

msg1 BYTE "REPORT HEADER", 0

msg2 BYTE "FOOTER", 0

msg3 BYTE "DEBUG SECTION", 0

.code

main PROC

mov EDX, OFFSET msg1

call WriteString

call Crlf

JMP Jumper

mov EDX, OFFSET msg3

call WriteString

call Crlf

Jumper:

mov EDX, OFFSET msg2

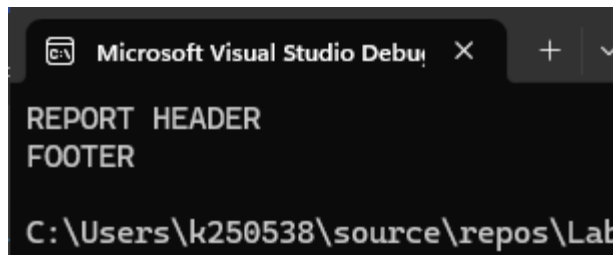
call WriteString

call Crlf

exit

main ENDP

END main



Task 5:

```
INCLUDE Irvine32.inc
```

```
.data
```

```
msg1 BYTE " X ", 0
```

```
msg2 BYTE " = ", 0
```

```
multiplier DWORD 1
```

```
outer DWORD ?
```

```
.code
```

```
main PROC
```

```
mov ECX, 3
```

```
L1:
```

```
mov outer, ECX
```

```
mov ESI, 1
```

```
mov ECX, 5
```

```
L2:
```

```
mov EAX, multiplier ;printing the first number
```

```
call WriteDec
```

```
mov EDX, OFFSET msg1 ;printing X
```

```
call WriteString
mov EAX, ESI ;printing the second number
call WriteDec
mov EDX, OFFSET msg2 ;printing =
call WriteString
mov EAX, multiplier
imul EAX, ESI ;multiplying the 1st and 2nd number
call WriteDec
```

```
inc ESI
call Crlf
loop L2
```

```
call Crlf
call Crlf
```

```
inc multiplier
mov ECX, outer
loop L1
```

```
exit
main ENDP
END main
```

```
Microsoft Visual Studio Debug Console
1 X 1 = 1
1 X 2 = 2
1 X 3 = 3
1 X 4 = 4
1 X 5 = 5

2 X 1 = 2
2 X 2 = 4
2 X 3 = 6
2 X 4 = 8
2 X 5 = 10

3 X 1 = 3
3 X 2 = 6
3 X 3 = 9
3 X 4 = 12
3 X 5 = 15

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```